

STRONG CHORDALITY OF GRAPHS WITH POSSIBLE LOOPS*

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Abstract. We unify two popular graph classes, strongly chordal graphs and chordal bigraphs, by introducing an umbrella class that contains both classes and maintains their essential properties. This is done by allowing loops at vertices. Considering loops often has little impact on a class of graphs; it however makes a big difference in this case. We call the new class *strongly chordal graphs with possible loops*. When all vertices have loops, we recover the usual strongly chordal graphs; when all vertices are loopless, we obtain the usual chordal bigraphs. Moreover, there is a surprising wealth of graphs in the new class that have loops at some vertices and not at others. These graphs also admit the elegant algorithms previously only applied in the extreme two cases. Formulated in the language of adjacency matrices, we study the class of symmetric 0,1 matrices that admit a simultaneous row and column permutation avoiding the Γ matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. We give ordering characterizations, matrix characterizations, and forbidden subgraph characterizations of the new class, and illustrate its usefulness by solving the minimum domination problem in this general context. This implies solutions of both the minimum dominating set in strongly chordal graphs and the minimum total dominating set in chordal bigraphs.

Key words. strongly chordal graph, totally balanced matrix, Gamma-free matrix, forbidden subgraph, domination, total domination

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1. Background and definitions. Strongly chordal graphs are one of the most popular graph classes, lying strictly between the class of interval graphs and the class of chordal graphs. There are problems that are intractable on chordal graphs but tractable on strongly chordal graphs, for instance, the problem of minimum dominating set. Chordal bipartite graphs (also known as chordal bigraphs) are a popular bipartite analogue of chordal graphs, characterized, like chordal graphs, by forbidding induced cycles. There are many natural analogies between strongly chordal graphs and chordal bigraphs, that arise from similar ordering (or matrix) characterizations. There are also natural transformations between the two classes.

We point out that if we consider the possibility of vertices having loops, both strongly chordal graphs and chordal bigraphs become special cases of one umbrella graph class, whose members we call *strongly chordal graphs with possible loops*. Indeed, strongly chordal graphs are best viewed as reflexive graphs (having loops at all vertices); their matrix characterizations typically add 1's to the main diagonal to state the results. Chordal bipartite graphs on the other hand, are best viewed as irreflexive (loop-free) graphs. Stated in this context, the ordering (matrix) characterization of

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these two classes becomes identical. The definition of strongly chordal graphs with possible loops makes use of this common ordering (matrix) characterization, which is in fact applicable to any graph with possible loops. We remark that other unifications of strongly chordal graphs and chordal bipartite graphs based on ordering (matrix) characterizations have been investigated, e.g., [4].

While loops are often irrelevant for graph concepts, we find that in this case they lead us to a new graph class that has interesting properties. We prove a forbidden induced subgraph characterization of strongly chordal graphs with possible loops, which includes all the forbidden subgraphs for reflexive strongly chordal graphs and all the forbidden subgraph for irreflexive chordal bigraphs, as well as additional graphs with some loops. We show that the adjacency matrices of strongly chordal graphs with possible loops correspond exactly to symmetric totally balanced matrices. (This result was known for matrices with an all-1 diagonal.) In particular, this class can be efficiently recognized.

As an illustration of the usefulness of this unified point of view, we revisit the minimum dominating set problem. The domination problem for graphs with possible loops encompasses both the usual domination problem (where every vertex dominates itself, i.e., the graph is reflexive), and the total domination problem (where every vertex has to be dominated by another vertex, i.e., the graph is irreflexive). We present a linear time certifying algorithm to solve this general domination problem for graphs with possible loops. This simultaneously solves the minimum dominating set problem and the minimum total dominating set problem.

We will say that a vertex is *reflexive* if it has a loop and *irreflexive* if it does not. A graph is *reflexive* if every vertex is reflexive and is *irreflexive* if every vertex is irreflexive. Thus the diagonal entries of the adjacency matrix of a reflexive graph are all 1, and those of an irreflexive graph are all 0. A *graph with possible loops* has in general a mixture of reflexive and irreflexive vertices, i.e., an arbitrary 0,1 diagonal of the adjacency matrix. In the figures, we will depict reflexive vertices in *black* and irreflexive vertices in *white*.

A reflexive graph G is *chordal* if it admits a *perfect elimination ordering* (also known as a *simplicial ordering*), $v_1, v_2, \dots, v_n$ of its vertices so that if $i < j, i < k$ and $v_i v_j \in E(G), v_i v_k \in E(G)$, then we also have $v_j v_k \in E(G)$. Chordal graphs are also characterized as those reflexive graphs that have no induced reflexive cycle of length greater than three, or those graphs that are intersection graphs of nonempty subtrees of a tree [9]. Note that we interpret the notion of intersection graph in a strict sense, where each nonempty set intersects itself, corresponding to the loops at all vertices. Chordal graphs admit efficient recognition algorithms and efficient algorithms for basic combinatorial problems that are intractable in general [9]. Note that a perfect elimination ordering corresponds to a simultaneous permutation of the rows and columns of the adjacency matrix that avoids the Γ matrix $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ as a principal submatrix. (Such a submatrix is called a *principal submatrix* if the upper left entry 1 lies on the main diagonal.)

A reflexive graph G is *strongly chordal* if it is chordal and if its adjacency matrix admits a simultaneous permutation of rows and columns that avoids the Γ matrix as any submatrix (not just as a principal submatrix). Strongly chordal graphs are equivalently characterized as those reflexive graphs that have no induced reflexive trampolines or induced reflexive cycles of length greater than three [6, 2]. They also admit efficient recognition algorithms [22], and efficient algorithms for some problems that are intractable for chordal graphs [7]; cf. the last section of this paper. A

geometric representation characterization of strongly chordal graphs has also been proposed [5].

Permuting rows and columns of a 0,1 matrix M to avoid Γ as a submatrix has been much studied [1, 17, 21, 22]. A Γ -free ordering of M is a matrix obtained from M by *independently* permuting its rows and columns, to avoid Γ as a submatrix. A *cycle matrix* is a square 0,1 matrix of size at least 3, with exactly two 1's in each row and each column, and no proper submatrix has the same property. A matrix is *totally balanced* if it has no cycle matrix as a submatrix [1, 17, 20]. If the constraint matrix of a linear program is totally balanced, then it can be solved by a greedy algorithm [1, 17]. A 0,1 matrix has a Γ -free ordering if and only if it is totally balanced [1, 17]. A symmetric 0,1 matrix with an all-1 diagonal has a symmetric Γ -free ordering if and only if it is totally balanced [21, 22]. We extend the latter result to all symmetric 0,1 matrices, regardless of the diagonal. Finding a Γ -free ordering (or deciding that one does not exist) admits efficient algorithms [21, 22]. Avoiding other submatrices has also been studied extensively; see, e.g., [19].

An arbitrary 0,1 matrix can be interpreted as the biadjacency matrix of a bigraph. Given a bigraph G (a bipartite graph with a fixed bipartition into red and blue vertices), we define the *biadjacency matrix* $N(G)$, with rows indexed by the red vertices and columns indexed by the blue vertices, and $N(i, j) = 1$ if and only if the i th red vertex is adjacent to the j th blue vertex. Note that $N(G)$ is in general not a square matrix, and any 0,1 matrix is the biadjacency matrix of some bigraph. A bigraph G is a *chordal bigraph* if its biadjacency matrix has a Γ -free ordering [11]. A geometric representation characterization of chordal bigraphs is given in [18].

For a square 0,1 matrix M , a *symmetric Γ -free ordering* is a *simultaneous* permutation of rows and columns that avoids Γ as a submatrix. Thus a reflexive graph G is strongly chordal if and only if its adjacency matrix $M(G)$ has a symmetric Γ -free ordering. The algorithm in [22] finds a symmetric Γ -free ordering of a symmetric matrix M (or decides that one does not exist) provided M has 1's on the main diagonal. We now consider the situation where the last restriction is lifted; i.e., we seek a symmetric Γ -free ordering of a symmetric 0,1 matrix M with arbitrary diagonal.

We define a graph G with possible loops to be *strongly chordal* if its adjacency matrix $M(G)$ admits a symmetric Γ -free ordering. Equivalently, a graph G with possible loops is strongly chordal if and only if its vertices can be linearly ordered as $v_1, v_2, \dots, v_n$ so that if $i < j, k < \ell$ and $v_i v_k \in E(G), v_i v_\ell \in E(G), v_j v_k \in E(G)$ (where i, j, k, ℓ are not necessarily all distinct), then we also have $v_j v_\ell \in E(G)$. This arises by straight translation of a symmetric Γ -free ordering of the adjacency matrix. We call such an ordering a *strong ordering* of G . A vertex $v \in V(G)$ is *simple* if its neighbors have their neighborhoods linear ordered by inclusion; i.e., if for any $x, y \in N(v)$, we have $N(x) \subseteq N(y)$ or $N(x) \supseteq N(y)$. A *simple ordering* of G is a linear ordering $v_1, v_2, \dots, v_n$ of $V(G)$ such that each v_i is simple in the subgraph induced by the set $\{v_i, v_{i+1}, \dots, v_n\}$. It is easy to see that a strong ordering is a simple ordering. We will prove that the converse also holds. These notions and facts are analogous to the usual theory of reflexive strongly chordal graphs [6], except we use open neighborhoods, and the neighborhood of a vertex w may or may not involve w , depending on whether w is reflexive or not. In particular, we see that for reflexive graphs this new notion of strong chordality coincides with the classical notion [6, 2].

Chordal bigraphs can be viewed as irreflexive strongly chordal graphs with possible loops. Indeed, consider the usual adjacency matrix $M(G)$ of a bipartite graph G , ordered so that all red vertices precede all blue vertices. Then it is clear that the

biadjacency matrix $N(G)$ is in the upper right corner of $M(G)$, and its transpose is in the lower left corner of $M(G)$. Moreover, an independent permutation of rows and columns of $N(G)$ yields a symmetric permutation of $M(G)$, and the biadjacency matrix $N(G)$ has a Γ -free ordering if and only if the adjacency matrix $M(G)$ has a symmetric Γ -free ordering.

Thus strongly chordal graphs with possible loops generalize both reflexive strongly chordal graphs and irreflexive chordal bigraphs. It follows from Theorem 3.1 that the new class coincides with the previously studied strongly chordal graphs G when G is reflexive, and coincides with the previously studied class of chordal bigraphs G when G is irreflexive.

It is useful to interpret the other definitions on the class of chordal bigraphs. Since a bigraph G is always irreflexive, a strong ordering of G corresponds to an ordering of the red vertices and an ordering of the blue vertices so that for red v_i, v_j and blue v_k, v_ℓ we have $i < j, k < \ell$, and $v_i v_k \in E(G), v_i v_\ell \in E(G), v_j v_k \in E(G)$ imply $v_j v_\ell \in E(G)$. A bigraph has a strong ordering if and only if it has a simple ordering [11]. A bigraph is chordal if and only if it does not contain an induced even cycle of length greater than four [10]. A bigraph G is chordal if and only if its biadjacency matrix $N(G)$ is totally balanced [1, 11, 17]. A geometric representation of chordal bigraphs has also been proposed [18].

In a companion paper [14] we define a digraph D to be a *strongly chordal digraph* if its adjacency matrix $M(D)$ admits a symmetric Γ -free ordering. It follows that a strongly chordal graph with possible loops is precisely (the underlying graph of) a symmetric strongly chordal digraph. It also follows that strongly chordal digraphs are chordal digraphs as defined in [12, 23]. The problem of recognizing strongly chordal digraphs is equivalent to the problem of deciding if a given square 0,1 matrix has a symmetric Γ -free ordering. This seems to be a difficult problem; as we show in [14], it is no longer equivalent with being totally balanced or any of the other polynomial conditions that applied for symmetric matrices with 1's on the main diagonal.

While graph classes are typically considered only for reflexive, or only for irreflexive graphs, other useful generalizations have been considered for graphs with possible loops. For example, in [16], the authors consider graphs with possible loops whose adjacency matrices avoid the submatrices $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ and find them to be a useful unification of interval graphs, adjusted interval digraphs, two-dimensional orthogonal ray graphs (alias interval containment digraphs), and complements of threshold tolerance graphs [8, 24, 15].

2. Extending classical characterizations. We prove the following extension of a result of Farber [6], who proved it for reflexive graphs. We will show in [14] that such results do not hold for digraphs or even for tournaments.

THEOREM 2.1. *Let G be a graph with possible loops. The following statements are equivalent:*

1. G is strongly chordal;
2. $M(G)$ is totally balanced;
3. every induced subgraph of G has a simple vertex;
4. G has a simple ordering.

Proof. If G is strongly chordal, it has a strong and hence a simple ordering. Consider the bigraph $B(G)$ obtained from G by replacing each vertex v by two vertices v_1, v_2 , and each edge vw by the two edges $v_1 w_2, w_1, v_2$. It is easy to see that $B(G)$ also has a simple ordering, whence the biadjacency matrix $N(G)$ is totally

balanced. Since $M(G) = N(B(G))$, this implies that $M(G)$ is totally balanced. Thus 1 implies 2.

To show 2 implies 3, suppose that $M(G)$ is totally balanced. Since every induced subgraph of a strongly chordal graph is obviously strongly chordal, it suffices to show that G has a simple vertex. Since $M(G) = N(B(G))$ is totally balanced, $B(G)$ is a chordal bigraph and hence has a simple vertex v_1 or v_2 for some vertex v of G , whence v is a simple vertex in G .

We will now show that 3 implies 1. So assume that every induced subgraph of G has a simple vertex. We show how to obtain a strong ordering $v_1, v_2, \dots, v_n$ of G . The selection of v_i for each $i \geq 1$ will be guided by a partial order $\preceq_i$ defined on $V_i = V(G) \setminus \{v_1, v_2, \dots, v_{i-1}\}$. Initially, $V_0 = V(G)$ and $\preceq_0$ on V_0 consists of the reflexive pairs only, that is, $x \preceq_0 y$ if and only if $x = y$ for all $x, y \in V_0$. For each $i \geq 1$, let $\preceq_i$ on V_i be defined by $x \preceq_i y$ if and only if $x \preceq_{i-1} y$ or $N_i(x) \subset N_i(y)$ where $N_i(x)$ and $N_i(y)$ are the neighborhoods of x and y in the subgraph of G induced by V_i . Equivalently, for each $i \geq 1$, $x \preceq_i y$ if and only if $x = y$ or $x \neq y$ and $N_j(x) \subset N_j(y)$ for some $j \leq i$. We will show that $\preceq_i$ is a partial order for each $i \geq 0$. The vertex v_i for each $i \geq 1$ is selected to be a simple vertex that is also a minimal element in the poset $(V_i, \preceq_i)$. We will also show that such a vertex v_i always exists.

First we prove that $\preceq_i$ is a partial order on V_i for each $i \geq 0$ by induction. Clearly, $\preceq_0$ is a partial order on V_0 . Assume that $i \geq 1$ and $\preceq_j$ is a partial order for each $j < i$. The reflexivity of $\preceq_i$ follows from the fact that $\preceq_i$ contains $\preceq_0$, which is reflexive. Suppose that $x \preceq_i y$ where $x \neq y$. Then there exists j with $j \leq i$ such that $N_j(x) \subset N_j(y)$. Thus $N_j(y) \not\subset N_j(x)$ for all $j \leq i$, i.e., $y \not\preceq_i x$. Hence $\preceq_i$ is antisymmetric. For the transitivity, suppose that $x \preceq_i y \preceq_i z$. Then there exist j, k with $j \leq i$ and $k \leq i$ such that $N_j(x) \subset N_j(y)$ and $N_k(y) \subset N_k(z)$. Let $\ell = \max\{j, k\}$. Then $\ell \leq i$ and $N_\ell(x) \subset N_\ell(z)$, which means that $x \preceq_i z$. Therefore $\preceq_i$ is a partial order on V_i for each $i \geq 0$.

Let u be a simple vertex in the subgraph of G induced by V_i . Such a vertex exists because every induced subgraph of G has a simple vertex. We prove that if $v \preceq_i u$, then v is also a simple vertex. So suppose that $v \preceq_i u$. Then there exists a j with $j \leq i$ such that $N_j(v) \subset N_j(u)$. Hence we must have $N_i(v) \subseteq N_i(u)$. Since u is simple, v is also simple. It follows that the subgraph of G induced by V_i has a simple vertex that is also a minimal element in the poset $(V_i, \preceq_i)$ for each $i \geq 1$. Therefore we obtain an ordering $v_1, v_2, \dots, v_n$. It suffices to show that the ordering is a strong ordering of G .

Suppose that $i < j$, $k < \ell$, $v_i \in N(v_k)$, $v_i \in N(v_\ell)$ and $v_j \in N(v_k)$. We show that $v_j \in N(v_\ell)$. By symmetry, we may assume that $i \leq k$. Thus $v_i, v_j, v_k, v_\ell \in V_i$. Since v_i is simple, either $N_i(v_k) \subseteq N_i(v_\ell)$ or $N_i(v_\ell) \subset N_i(v_k)$. In the latter case, $v_\ell \preceq_i v_k$, and hence $v_\ell \preceq_k v_k$. However, v_k is minimal in $(V_k, \preceq_k)$ by the choice of v_k , a contradiction. Therefore $N_i(v_k) \subseteq N_i(v_\ell)$, which implies $v_j \in N(v_\ell)$. This shows that the ordering $v_1, v_2, \dots, v_n$ is a strong ordering of G .

Finally we note that statements 3 and 4 are obviously equivalent. $\square$

Let $W : v_0 v_1 \dots v_k$ be a walk of length k in a graph G with possible loops. If $v_0 = v_k$, then W is called a *closed walk*. A *subwalk* of a walk W is a walk $v_i v_{i+1} \dots v_j$ for some $0 \leq i \leq j \leq k$. A subwalk of W is *proper* if the length of the subwalk is less than the length of W . A *strong chord* of a walk W is an edge $v_i v_j$ (possibly a loop if $v_i = v_j$) such that $j - i$ is odd but not equal to 1 or -1 . A *strong chord of a closed walk* W is defined similarly, except the expression $j - i$ is evaluated modulo k . Note that in a closed walk $W : v_0, v_1, \dots, v_k = v_0$, the last edge $v_{k-1} v_0$ is not an strong chord for any k because 0 is $(k - 1) + 1$ modulo k .

COROLLARY 2.2. *A graph G with possible loops is strongly chordal if and only if every even closed walk of length at least 6 has a strong chord [1].* $\square$

Proof. This follows from Theorem 2.1 and the fact $M(G)$ is totally balanced if and only if in G every even closed walk of length at least 6 has a strong chord [1]. $\square$

A matrix reformulation of this result states the following: *a symmetric $0, 1$ matrix which has a Γ -free ordering also has a symmetric Γ -free ordering.* For matrices with 1's on the main diagonal, this was proved in [22].

3. Forbidden induced subgraphs. Corollary 2.2 characterizes strongly chordal graphs with possible loops by means of a forbidden structure, namely, even closed walks without strong chords. Recall that in the reflexive case, a characterization is known in terms of forbidden induced subgraphs, namely, cycles of length at least four, and trampolines [6] (also called suns [2]). In the irreflexive case, it turns out that a characterization by forbidden induced subgraphs is also known. First we note that all odd cycles are forbidden, because going around an odd cycle twice produces an even closed walk without strong chords. Thus, for irreflexive graphs only bipartite graphs can have a Γ -free ordering, and the characterization from [10] gives the forbidden induced subgraphs of chordal bigraphs, namely, all even cycles of length greater than four. In conclusion, for irreflexive graphs, the forbidden induced subgraphs are all cycles of length different from four, see the bottom-left sequence in Figure 1. The reflexive cycles of length at least four shown in the top-left sequence in Figure 1, together with the trampolines included in Figure 3, comprise the forbidden induced subgraphs for reflexive graphs. Figure 1 also contains additional forbidden induced cycles of length at least five with either one or two loops.

Next we consider obstructions that are neither reflexive nor irreflexive. Assume C is a cycle with vertices $0, 1, \dots, n-1, n > 4$, with one loop, at 0; or two loops, at 0 and $n-1$. A *regular fan at 0* is the set of edges $0i$ for all even subscripts i . A *regular fan at $n-1$* is the set of edges $(n-1)(n-j)$ for all odd subscripts j . These concepts are illustrated in Figure 2.

There is an infinite family of forbidden induced subgraphs consisting of even cycles with a loop and regular fan at 0 as on the left side of Figure 2, and two families of cycles with loops at $0, n-1$ – one with a regular fan only at 0, illustrated on the upper right of Figure 2, and one with regular fans both at 0 and at $n-1$, as on the lower right of Figure 2. It can be readily checked that each of these graphs contains an even closed walk of length at least 6 without strong chords. Another infinite family of forbidden

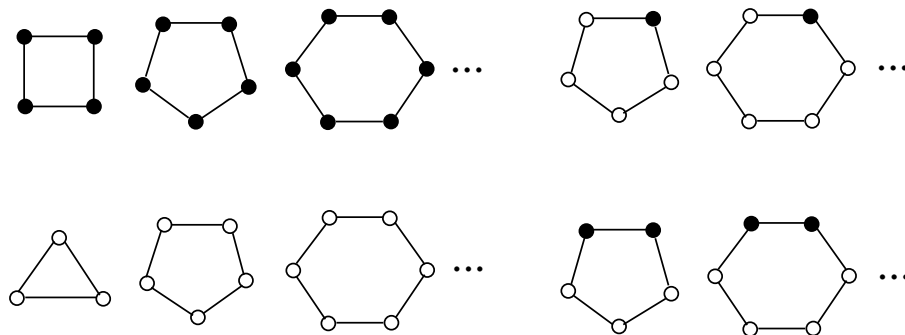
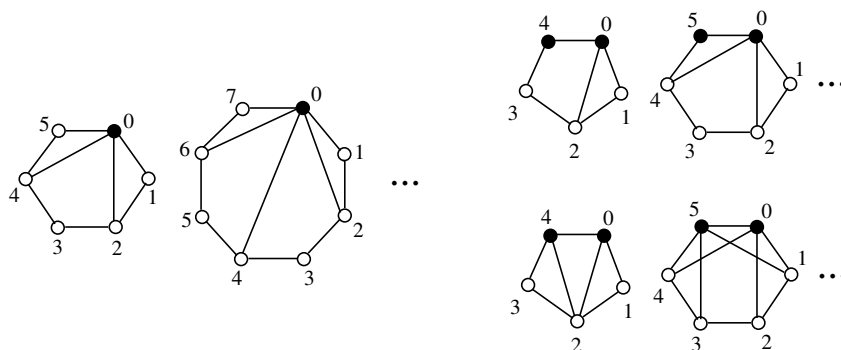
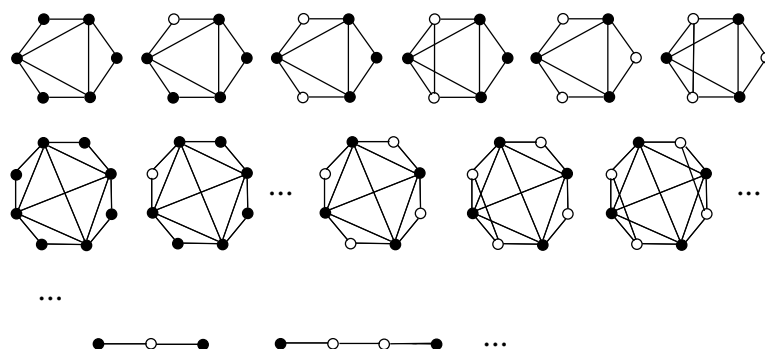


FIG. 1. Forbidden chordless cycles: families $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4$

FIG. 2. Forbidden cycles with fans: families $\mathcal{F}_5, \mathcal{F}_6, \mathcal{F}_7$ FIG. 3. Forbidden paths and weak trampolines: families $\mathcal{F}_8, \mathcal{F}_9$

induced subgraphs consists of *weak trampolines*; these are obtained from reflexive trampolines by removing loops from an arbitrary subset S of the vertices of degree 2, and adding an arbitrary set of disjoint edges between pairs of the vertices in S ; cf. Figure 3. (Note that this definition includes trampolines in the classical sense.) Finally, any path joining two reflexive vertices by a sequence of irreflexive vertices is also a forbidden induced subgraph, illustrated on the bottom of Figure 3. In each of these graphs one can find an even closed walk of length at least 6 without strong chords.

The following more precise description characterizes strongly chordal graphs with possible loops by forbidden induced subgraphs.

THEOREM 3.1. *A graph G with possible loops is strongly chordal if and only if it does not contain as an induced subgraph a graph in any of the following families:*

1. Family $\mathcal{F}_1$: reflexive cycles of length at least 4;
2. Family $\mathcal{F}_2$: irreflexive cycles of length other than 4;
3. Family $\mathcal{F}_3$: cycles of length at least 5 with exactly one loop;
4. Family $\mathcal{F}_4$: cycles of length at least 5 with exactly two consecutive loops;
5. Family $\mathcal{F}_5$: even cycles of length at least 6 with a loop at 0, with a regular fan at 0;
6. Family $\mathcal{F}_6$: cycles of length at least 5 with two loops, at 0 and $n - 1$, with a regular fan at 0;
7. Family $\mathcal{F}_7$: cycles of length at least 5 with two adjacent loops, at 0 and $n - 1$, and regular fans at both 0 and $n - 1$;
8. Family $\mathcal{F}_8$: weak trampolines; and
9. Family $\mathcal{F}_9$: paths of length at least 2 with two loops at the two end vertices.

We write $\mathcal{F} = \bigcup_{i=1}^9 \mathcal{F}_i$. As we noted, every graph in $\mathcal{F}$ contains an even closed walk of length at least 6, without strong chords. Thus by Corollary 2.2, we conclude that any graph with possible loops that contains a graph from $\mathcal{F}$ as an induced subgraph is not strongly chordal. We prove the converse of this statement is also true.

The following two lemmas describe the cases of reflexive and irreflexive graphs, and follow from known results on chordal bipartite graphs [9] and strongly chordal graphs [6], as discussed above.

LEMMA 3.2. *If the subgraph of G induced by reflexive vertices is not a chordal graph, then G contains a graph in $\mathcal{F}_1$ as an induced subgraph. If the subgraph of G induced by irreflexive vertices is not a chordal bigraph, then G contains a graph in $\mathcal{F}_2$ as an induced subgraph.*

LEMMA 3.3. *Suppose that H is a chordal graph on vertices $v_0, v_1, \dots, v_{k-1}$ where $k \geq 6$ is even. If $N(v_i) = \{v_{i-1}, v_{i+1}\}$ for each even i and the vertices with even subscripts form an independent set, then H contains a graph in $\mathcal{F}_8$ as an induced subgraph. In particular, if a chordal graph contains an even closed walk of length at least 6 without strong chords, then it contains a graph from $\mathcal{F}_8$ as an induced subgraph.*

Suppose $W : v_0, v_1, \dots, v_j = v_0, v_{j+1}, \dots, v_k$ is any walk, and consider its closed proper subwalk $W' : v_0, v_1, \dots, v_j (= v_0)$. If j is even, then the edge $v_j v_{j+1}$ is a strong chord, and if j is odd and v_0 has a loop, then that loop is a strong chord. For future use, we formalize these observations as follows.

LEMMA 3.4. *Suppose $W : v_0, v_1, \dots, v_j = v_0, v_{j+1}, \dots, v_k$ is a walk without strong chords. Then its closed proper subwalk $W' : v_0, v_1, \dots, v_j (= v_0)$ has an odd length. Moreover, the vertex v_0 must be irreflexive, unless $W = W' v_0$.*

Note that in particular a closed walk of odd length that has no strong chords can only self-intersect if it uses a loop; i.e., if $W : v_0, v_1, \dots, v_j = v_0, v_{j+1}, \dots, v_k = v_0$ is a closed walk without strong chords, and k is odd, then $j = 1$ or $j + 1 = k$.

The next two auxiliary lemmas describe the possible shape of cycles with exactly one or two loops in a graph with possible loops which does not contain an induced subgraph from $\mathcal{F}$. They will be used repeatedly in our arguments.

LEMMA 3.5. *Let G be a graph with possible loops, and let $C : v_0 v_1 \dots v_k$, $k \geq 5$, be a cycle in G , where $v_0 = v_k$ is the only reflexive vertex of C . Suppose moreover, that the subpath $v_0 v_1 \dots v_{k-1}$ of C has no strong chords. Then k is odd and $v_0 v_j \in E(G)$ for all even $j < k$, or G contains an induced subgraph from $\mathcal{F}_2 \cup \mathcal{F}_3 \cup \mathcal{F}_5$.*

Proof. We first note that the subpath $v_1 v_2 \dots v_{k-1}$ is an induced path; otherwise it would have a chord, and since the chord is not strong, G would contain an induced irreflexive cycle of length other than 4, i.e., a graph from $\mathcal{F}_2$. Moreover, $v_0 v_i \notin E(G)$ for each odd i with $1 < i < k - 1$, as these edges would be strong chords. Suppose for contradiction that $v_0 v_j \notin E(G)$ for some even j . Let ℓ be the greatest subscript with $1 \leq \ell < j$ such that $v_0 v_\ell \in E(G)$ and r be the least subscript with $j < r \leq k - 1$ such that $v_0 v_r \in E(G)$. Then $v_0, v_\ell, v_{\ell+1}, \dots, v_r$ induce a cycle in $\mathcal{F}_3$. Hence $v_0 v_j \in E(G)$ for each even j ; furthermore, k must be odd, as otherwise G would contain a cycle in $\mathcal{F}_5$. $\square$

LEMMA 3.6. *Suppose G is a graph with possible loops and $W : v_0 v_1 \dots v_k$ is a closed walk in G , of odd length $k > 3$, such that the subwalk $v_0 v_1 \dots v_{k-1}$ has no strong chords. Suppose moreover that v_0 and v_{k-1} are the only reflexive vertices in W . Then G contains an induced subgraph from $\bigcup_{i=2}^7 \mathcal{F}_i$.*

Recall that a strong chord in a walk not viewed as closed is an edge $v_i v_j$ with $j - i$ odd and not equal to 1, -1 , where the difference is not computed modulo k .

Proof. We first observe that we may assume that W is a cycle; indeed Lemma 3.4 specifies any proper subwalk would be odd, which is not possible for an odd walk. (Note that the last option $W = vW'$ would imply that k is even, so it cannot occur.)

Since W has no strong chords, we must have $v_0v_i \notin E(G)$ for each odd i with $1 < i < k$ and $v_iv_{k-1} \notin E(G)$ for each odd i with $0 < i < k-2$. Let ℓ be the greatest subscript with $1 \leq \ell < k-1$ such that $v_0v_\ell \in E(G)$, and let r be least subscript with $0 < r \leq k-2$ such that $v_rv_{k-1} \in E(G)$. If $\ell < r$, then $v_0, v_\ell, v_{\ell+1}, \dots, v_r, v_{k-1}$ induce a graph in $\mathcal{F}_4$. So assume that $\ell \geq r$. Note that ℓ and r are both even. Applying Lemma 3.5 to the cycle $v_0v_1 \dots v_\ell v_0$ we may conclude that $v_1v_j \in E(G)$ for each even j with $0 < j \leq \ell$ (else G contains an induced subgraph from $\mathcal{F}_2 \cup \mathcal{F}_3 \cup \mathcal{F}_5$). Similarly, applying Lemma 3.5 to the cycle $v_{k-1}v_{k-2} \dots v_rv_{k-1}$ we may conclude that $v_jv_{k-1} \in E(G)$ for each even j with $r \leq j < k-1$. If $\ell = k-3$ and $r = 2$, then $v_0, v_1, \dots, v_{k-1}$ would induce a graph in $\mathcal{F}_7$ as an induced subgraph. So we must have $\ell < k-3$ or $r > 2$, and so $v_\ell, v_{\ell+1}, \dots, v_{k-1}, v_0$ or $v_{k-1}, v_0, v_1, \dots, v_r$ induce a graph in $\mathcal{F}_6$. $\square$

LEMMA 3.7. *Suppose that u, v are nonadjacent reflexive vertices in G . If there is an induced (u, v) -path whose internal vertices are not all reflexive, then G contains a graph from $\mathcal{F}_9$ as an induced subgraph. In particular, if there is a (u, v) -walk whose internal vertices are all irreflexive, then G contains a graph from $\mathcal{F}_9$ as an induced subgraph.*

We are now ready to prove the missing direction for Theorem 3.1.

LEMMA 3.8. *If G is not strongly chordal, then it contains a graph in $\mathcal{F}$ as an induced subgraph.*

Proof. Suppose that G is not strongly chordal. By Corollary 2.2, G contains an even closed walk $W : v_0v_1 \dots v_k$, of length at least 6, without strong chords. Consider first the case when W is not a cycle (i.e., W contains a repeated vertex).

Suppose first that there is a vertex which appears twice consecutively in W , say, $v_0 = v_1$. (Thus v_0 is a reflexive vertex, and the loop v_0v_0 is an edge of W .) Since W has no strong chords, $v_0v_i = v_1v_i \notin E(G)$ for any $2 < i < k-1$. We claim that $v_i \neq v_0$ for any $1 < i < k$. Indeed, if $v_i = v_0$, then $v_1v_2 \dots v_i$ is a proper closed walk of W . This contradicts Lemma 3.4, as $v_1 = v_i$ is a reflexive vertex. The subwalk $W' : v_1v_2 \dots v_k$ is a closed walk of an odd length. If v_0 is the only reflexive vertex in W and W' is a cycle, then by Lemma 3.5 (applied to W') G contains a graph in $\mathcal{F}$ as an induced subgraph (since we have shown that $v_0v_i = v_1v_i \notin E(G)$).

If v_0 is the only reflexive vertex in W and W' is not a cycle, then W' contains a cycle not containing v_0 , which implies G contains a graph in $\mathcal{F}_2$ as an induced subgraph.

Suppose that W has exactly two reflexive vertices. Let v_a be the other reflexive vertex in W . If $a \notin \{2, k-1\}$, then $v_0v_a \notin E(G)$ as otherwise W contains the strong chord $v_0v_a = v_1v_a$, a contradiction to the assumption that W has no strong chords. Thus $v_0v_1 \dots v_a$ is a (v_0, v_a) -walk whose internal vertices are all irreflexive. By Lemma 3.7, G contains a graph in $\mathcal{F}_9$ as an induced subgraph. So $a \in \{2, k-1\}$. If $a = k-1$, then by Lemma 3.6 (applied to $v_1v_2 \dots v_k$) G contains a graph in $\bigcup_{i=2}^7 \mathcal{F}_i$ as an induced subgraph. If $a = 2$, then again by Lemma 3.6 (applied to $v_1v_{k-1}v_{k-2} \dots v_1$) G contains a graph in $\bigcup_{i=2}^7 \mathcal{F}_i$ as an induced subgraph.

Suppose now that W has more than two reflexive vertices. A similar proof as above shows that v_2 and v_{k-1} are both reflexive. Since W does not contain a strong chord, $v_2v_{k-1} \notin E(G)$. Let $v_2v_{j_1} \dots v_{j_t}v_{k-1}$ be the shortest (v_2, v_{k-1}) -path in the

subgraph of G induced by $V(W - v_0)$. If some v_{j_i} is irreflexive, then G contains a graph in $\mathcal{F}_9$ as an induced subgraph according to Lemma 3.7. On the other hand if each v_{j_i} is reflexive, then $v_0, v_2, v_{j_1}, \dots, v_{j_t}, v_{k-1}$ induce a graph in $\mathcal{F}_1$. This completes the case when a vertex appears twice consecutively in W , i.e., a loop is an edge of W .

Suppose next that W contains a repeated vertex but no vertex appears twice consecutively in W . Then W contains a closed proper walk. By Lemma 3.4 such a walk is of an odd length and no reflexive vertex can be a repeated vertex in W . Without loss of generality assume that $W' : v_0 v_1 \dots v_c$ is such a walk. Then c is odd and v_0 is irreflexive. Since c is odd, $W'' : v_c v_{c+1} \dots v_k$ is a closed proper subwalk of W of an odd length. If W' or W'' contains only irreflexive vertices, then G contains an odd cycle consisting of irreflexive vertex and hence a graph in $\mathcal{F}_2$ as an induced subgraph. So we may assume that W' and W'' both contain reflexive vertices. Let v_f be the reflexive vertex in W' with the greatest subscript and v_g be the reflexive vertex in W'' with the least subscript. The choice of v_f, v_g implies that the walk $v_f v_{f+1} \dots v_g$ whose internal vertices are all irreflexive. By Lemma 3.7, G contains a graph in $\mathcal{F}_9$ as an induced subgraph, or $v_f v_g \in E(G)$. So we may assume $v_f v_g \in E(G)$. Since W does not contain a strong chord, $g - f$ is even. If $f \neq c - 1$ or $g \neq c + 1$, then $v_f v_{f+1} \dots v_g v_f$ is a closed walk of an odd length > 3 without strong chords, in which v_f and v_g are the only reflexive vertices. Applying Lemma 3.6 to this walk, we conclude that G contains a graph in $\bigcup_{i=2}^7 \mathcal{F}_i$ as an induced subgraph. Hence we may assume that $f = c - 1$ and $g = c + 1$. Let $v_{f'}$ be the reflexive vertex in W' with the least subscript (possibly $f' = f$) and $v_{g'}$ be the reflexive vertex in W'' with the greatest subscript (possibly $g' = g$). By considering the walk $v_{g'} v_{g'+1} \dots v_k v_1 \dots v_{f'}$ and using a similar argument as for v_f, v_g , we may conclude that $v_{f'} v_{g'} \in E(G)$, and $f' = 1, g' = k - 1$. Since no reflexive vertex can be a repeated vertex, $v_{f'} \neq v_g$. Since W does not contain a strong chord, $v_{f'} v_g \notin E(G)$. Hence $v_{f'}, v_0, v_g$ induce a graph in $\mathcal{F}_9$.

Consider now the case when W is a cycle. In view of Lemmas 3.2, 3.3, 3.5, and 3.6, we assume that W contains an irreflexive vertex and at least three reflexive vertices. Suppose that W contains consecutive irreflexive vertices. Without loss of generality assume that $v_1, v_2, \dots, v_{h-1}$ are irreflexive vertices where $h > 2$ and that v_0 and v_h are reflexive. If $v_0 v_h \notin E(G)$, then $v_0 v_1 \dots v_h$ is a walk connecting two reflexive vertices whose internal vertices are all irreflexive. By Lemma 3.7, G contains a graph in $\mathcal{F}_9$ as an induced subgraph. So assume that $v_0 v_h \in E(G)$. Then h is even as W has no strong chords. Applying Lemma 3.6 to the cycle $v_0 v_1 \dots v_h v_0$ we conclude that G contains a graph in $\bigcup_{i=2}^7 \mathcal{F}_i$ as an induced subgraph. Hence we assume that W contains no consecutive irreflexive vertices.

We prove by contradiction that any two irreflexive vertices are of an even distance from each other in W . So suppose that v_r, v_s are two irreflexive vertices whose distance in W is odd. Since the distance of v_r, v_s in W is odd, r, s have different parity. Since no consecutive vertices in W are irreflexive, the distance of v_r, v_s is at least 3. Assume without loss of generality that r is odd and s is even. If $v_{r-1} v_{r+1} \notin E(G)$, then $\{v_{r-1}, v_r, v_{r+1}\}$ induce a graph in $\mathcal{F}_9$. So assume $v_{r-1} v_{r+1} \in E(G)$. Similarly, we may assume that $v_{s-1} v_{s+1} \in E(G)$. Let $P : v_{r+1} v_{\alpha_1} \dots v_{\alpha_p} v_{s-1}$ be an induced (v_{r+1}, v_{s-1}) -path in the subgraph of G induced by $v_{r+1}, v_{r+2}, \dots, v_{s-1}$, and let $Q : v_{r-1} v_{\beta_1} \dots v_{\beta_q} v_{s+1}$ be an induced (v_{r-1}, v_{s+1}) -path in the subgraph of G induced by $\{v_{r-1}, v_{r-2}, \dots, v_{s+1}\}$. If any of P and Q contains an internal irreflexive vertex, then G contains a graph in $\mathcal{F}_9$ as an induced subgraph by Lemma 3.7. So assume that vertices in P and Q are all reflexive. The subgraph of G induced by $V(P) \cup V(Q)$ contains the reflexive cycle C formed by P, Q , and the edges $v_{r-1} v_{r+1}$ and $v_{s-1} v_{s+1}$.

Note that the subscripts $r-1, r+1$ are even and $s-1, s+1$ are odd; thus the path P starts with a vertex with even subscript and ends with a vertex with odd subscript (and similarly for Q). Thus the path P includes an edge $v_{\alpha_i}v_{\alpha_{i+1}}$ where the subscript α_i is even and the subscript α_{i+1} is odd. The only chords possible in C are between a vertex in P and a vertex in Q , as these paths are induced; moreover since there are no strong chords in W , the only chords possible in C are between vertices with subscripts of the same parity. Note that the edge $v_{\alpha_i}v_{\alpha_{i+1}}$ does belong to some cycle (e.g., C), but the shortest cycle it belongs to has length greater than three, as the vertices of Q with even subscripts are not adjacent to $v_{\alpha_{i+1}}$, and the vertices of Q with odd subscripts are not adjacent to v_{α_i} . Thus C induces a graph that is not chordal and contains a graph in $\mathcal{F}_1$ as an induced subgraph by Lemma 3.2. Therefore any two irreflexive vertices are of an even distance from each other in W .

Since W contains at least one irreflexive vertex, we may assume without loss of generality that v_0 is irreflexive. Then all irreflexive vertices in W have even subscripts. Suppose that there is no edge between any two vertices of even subscript (i.e., the vertices of even subscripts form an independent set). If v_i is irreflexive and $v_{i-1}v_{i+1} \notin E(G)$, then v_{i-1}, v_i, v_{i+1} induce a graph in $\mathcal{F}_9$. Thus we assume that for each irreflexive v_i , the two neighbors v_{i-1}, v_{i+1} of v_i are adjacent. If the subgraph of G induced by the reflexive vertices in W is not chordal, then G contains a graph in $\mathcal{F}_1$ as an induced subgraph. On the other hand if the subgraph of G induced by the reflexive vertices in W is chordal, then the subgraph of G induced by $V(W)$ is also chordal because each irreflexive vertex is simplicial in the subgraph. By Lemma 3.3, the subgraph of G induced by $V(W)$ contains a graph in $\mathcal{F}_8$ as an induced subgraph.

It remains to consider the case when there are edges between vertices with even subscripts. If there is an edge between an irreflexive vertex v_i and a reflexive vertex v_j with even j , then v_{i-1}, v_i, v_j induce a graph in $\mathcal{F}_9$, since W has no strong chords.

Assume first that v_bv_d is an edge between two reflexive vertices with even b, d , and assume that $0 < b < d$, and the difference $d-b$ is as large as possible. Consider any induced (v_{d+1}, v_{b-1}) -path P' in G contained in $\{v_{d+1}, v_{d+2}, \dots, v_{k-1}, v_0, v_1, \dots, v_{b-1}\}$. If any internal vertex of P' is irreflexive then G contains a graph from $\mathcal{F}_9$ as an induced subgraph, by Lemma 3.7. Thus P' together with the edges $v_{b-1}v_b, v_bv_d, v_dv_{d+1}$ forms a reflexive cycle containing the edge v_bv_d . By Lemma 3.2 we can assume that the reflexive vertices induce a chordal graph, and thus the edge v_bv_d must belong to a 3-cycle with some vertex v_c . Since b, d are even, c must also be even (else at least one of v_bv_c, v_dv_c is a strong chord of W). If $0 < c < b$, then $d-c > d-b$ which violates the choice of v_b, v_d . Similarly, if $d < c < k$, then also $c-b > d-b$ which also violates the choice of v_b, v_d .

Finally, we consider edges between two vertices of even subscripts when these two vertices are both irreflexive. If such edges form a matching in G then G contains a graph $\mathcal{F}_8$ as an induced subgraph. If these edges do not form a matching, then one can verify that G must contain a graph from $\mathcal{F}_6 \cup \mathcal{F}_7$ as an induced subgraph. This completes the proof of Theorem 3.1. $\square$

4. An example application. In this section we note that problems for graphs with possible loops are often very natural and combine the reflexive and irreflexive versions in interesting ways. Consider for example the problems of domination and total domination. The *domination number* of a graph G is the minimum number of vertices in a *dominating set* D , i.e., a set such that each vertex is in D or has a neighbor in D . The *total domination number* of a graph G is the minimum number of vertices in a *total dominating set* D , i.e., a set such that each vertex has a neighbor in

D . This suggests that the former deals with graphs that are reflexive, as each vertex dominates itself (as if it had a loop), while the latter deals with graphs that are irreflexive, no vertex is adjacent to itself, so it cannot dominate itself. We can more generally define the *general domination number of a graph G with possible loops* to be the minimum number of vertices in a set D such that each vertex has a neighbor in D . If G is reflexive, the general domination number coincides with the usual domination number, and if G is irreflexive, it coincides with the total domination number. For a general graph with possible loops, this new problem represents an interesting mixture of the two classical problems.

Farber [7] gave a linear time algorithm to find a minimum dominating set in a reflexive strongly chordal graph, and Damaschke, Mueller, and Kratsch [3] gave a linear time algorithm for a minimum total dominating set in a bipartite chordal graph. We now describe a linear time certifying algorithm for the problem to find a minimum general dominating set in a strongly chordal graph with possible loops, which generalizes and extends both the algorithms in [7, 3]. The algorithm follows the spirit of the algorithm of Farber [7] by simultaneously solving the problem and its linear programming dual. This illustrates just how natural the new class of strongly chordal graphs with possible loops is.

As in [7], we consider the minimum general dominating set problem together with its dual, the maximum number of vertices with disjoint neighborhoods. While in Farber's case, the neighborhoods were all closed neighborhoods, for us they are open neighborhoods, which may or may not include the vertex itself, depending on whether or not it has a loop. Clearly a vertex can be dominated only by a vertex from its neighborhood, so vertices with disjoint neighborhoods need to be dominated by distinct vertices. Therefore, the maximum number of vertices with disjoint neighborhoods is a lower bound for the minimum number of vertices in a dominating set. Moreover, if we find a dominating set D and a set C of vertices with disjoint neighborhoods such that $|C| = |D|$, then D is minimum and C is maximum.

These sets will be computed iteratively in the order of a strong ordering $<$ of a graph G with possible loops. We repeat the following steps, each assigning to some vertices of G labels C , D , and N . (The label N signals that the label C is no longer available for that vertex.) We assume that G has no isolated vertices, as such vertices can always be dealt with separately. Initially, no vertices are labeled.

- Find, in the ordering $<$, the first vertex x without the label N .
- Find, in the ordering $<$, the last neighbor y of x .
- Label x by C , label y by D , and label all neighbors of y by N .

Note that a vertex will in general receive several labels. Every vertex will receive at least the label C or N . Moreover, every vertex will receive the label C and D at most once. Specifically, when a vertex x is labeled C , a unique neighbor y of x is labeled D . At that point, all neighbors of y receive the label N , including x . Therefore x will not receive the label C again, and y will never be receiving another label D (since all its neighbors are ineligible for label C). So, if we ignore the auxiliary labels N , we will have some k vertices labeled C and the same number k vertices labeled D . (Some vertices may have both labels C and D .) In other words, we have sets C and D (of vertices with those labels) that have the same cardinality. The final set D is dominating, as there are no vertices left without a label C or N , and each vertex with label C or N has a neighbor labeled D . We now prove that the neighborhoods of vertices labeled C are disjoint. Otherwise some $x < x'$ both labeled C have a common neighbor z ; suppose y was the last neighbor of x when x was labeled C . Since y is the last neighbor of x , we have $z < y$. Since x' is labeled

C later than x , it is not a neighbor of y . Therefore we have $x < x', z < y$, and $xz \in E(G)$, $xy \in E(G)$, and $x'z \in E(G)$, $x'y \notin E(G)$, which contradicts the fact that $<$ is a strong ordering. Thus we have a general dominating set D and a set of vertices C with disjoint neighborhoods, and $|C| = |D|$. Therefore both are optimal.

We have given a linear time algorithm solving the general domination problem (and its dual) if a strong ordering of the graph with possible loops is given. We expect that the algorithm for weighted domination in strongly chordal graphs [7] also allows a similar extension to weighted general domination in strongly chordal graphs with possible loops.

5. Conclusions. We have defined strong chordality for graphs with possible loops and found them to behave in a way analogous to reflexive chordal graphs and to chordal bigraphs. In particular, they correspond to totally balanced matrices and hence can be efficiently recognized, they enjoy the equivalence of strong and simple orderings, and they admit an elegant forbidden induced subgraph characterization.

The unified perspective offers some advantages, as we illustrated on the problem of domination and total domination that can be handled together on the new graph class.

Strongly chordal digraphs can be defined by a similar mechanism, and we will do that in a companion paper [14]. However, the nice properties exhibited here mostly do not extend to strongly chordal digraphs. In [14] we discuss these facts and focus on strongly chordal digraphs in restricted classes of digraphs. The relevant definitions and some preliminary results can be found in [13].

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